

HESOD: HIGH-RESOLUTION EFFICIENT SMALL OBJECT DETECTION VIA SEMANTIC-SPECTRAL DUAL SELECTION

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ABSTRACT

Tiny-object detection in high-resolution aerial and maritime imagery must contend with vast, redundant backgrounds and extreme scale variation. While spatial selective computation routes foreground regions to cut latency, existing single-evidence selectors suffer severe semantic collapse on micro targets, causing irrecoverable pruning. We propose **HESOD** (High-Resolution Efficient Small Object Detection), a dual-evidence framework unifying semantic-spectral routing, coverage-constrained optimization, and lightweight detection. HESOD couples semantic objectness with constant-width spectral saliency under a monotonic evidence-preserving max-fusion rule to rescue degraded targets. An object-level soft-coverage loss optimizes multi-cell instance coverage across ground-truth objects for recall-safe routing. To recover downstream efficiency, an Inverse-Residual Shared-Parameter Partial-Convolution (ISPP) head compresses parameters by $>27\%$ with only fractional FLOPs overhead. Experiments show that HESOD improves mAP@.5 by 0.9 pp and 2.3 pp (boosting total instance recall by 3.66 pp and 3.33 pp) on UAVDT and SeaPerson, respectively, at up to 106 FPS, making it well suited to resource-constrained deployment.

Index Terms— Small object detection, selective computation, spectral saliency, coverage optimization, UAV

1. INTRODUCTION

High-resolution aerial and maritime vision is fundamental to intelligent surveillance, environmental monitoring, and search-and-rescue. In operational scenarios across aerial and maritime domains [1–3], targets of interest typically occupy only a handful of pixels ($<16 \times 16$ px) amid massive uniform backgrounds ($>70\%$ – 80%) [5]. Evaluating standard dense deep detectors across multi-megapixel grids (1280×1280 to 2048×2048) incurs prohibitive arithmetic latency on resource-constrained platforms.

To eliminate redundant computation, *spatial selective computation* routes feature processing to prospective foreground regions [6, 7]. Notably, ESOD [4] predicts an early spatial priority heatmap after a shallow stem, slicing features into candidate patches (AdaSlicer) and bypassing background across downstream stages.

However, spatial selection operates under a severe *asymmetric error cost*: false-positive background patches merely incur minor compute overhead, whereas discarding a true tiny target produces an *irrecoverable false negative*. Existing detectors rely on single-semantic objectness with independent per-cell BCE. Under extreme

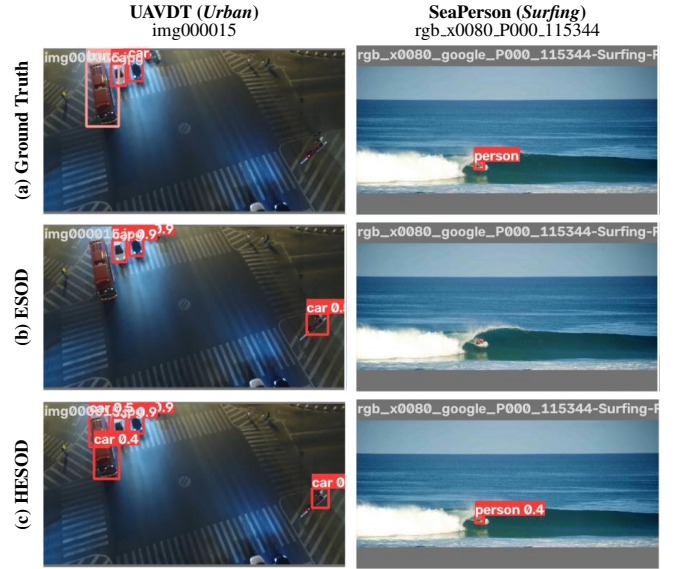


Fig. 1. Visual comparison on UAVDT and SeaPerson. Baseline ESOD drops micro targets (b), while HESOD restores full instance recall (c).

downsampling, micro targets suffer severe semantic collapse, causing faint instances to be prematurely pruned (Fig. 1(b)). To overcome this recall dilemma, we observe that tiny targets retain high-frequency spatial gradients and spectral saliency even when deep semantics fade [8–10]. We then propose High-Resolution Efficient Small Object Detection (**HESOD**) to establish recall-safe, evidence-preserving sparse routing. By coupling dual-evidence routing with an object-level soft-coverage loss, our selector reliably rescues these dropped targets (Fig. 1(c)). Our contributions include:

1. **Dual-Evidence Spatial Routing:** We uncover single-semantic routers’ vulnerability to feature collapse, coupling semantic objectness with constant-width gradient filtering for scale-robust structural priors independent of backbone capacity.
2. **Evidence Preservation and Soft Coverage:** We formulate a monotonic max-fusion rule preventing multi-cue evidence dilution, paired with an object-level noisy-OR soft-coverage loss directly optimizing joint instance recall.
3. **Compute Compensation and Empirical Gains:** We co-design an ISPP head that reduces detector parameters by $>27\%$ while improving mAP@.5 by 0.9/2.3 pp on UAVDT and SeaPerson, boosting total instance recall by 3.66/3.33 pp.

* Source code at <https://github.com/deepsota-ai/HESOD>.

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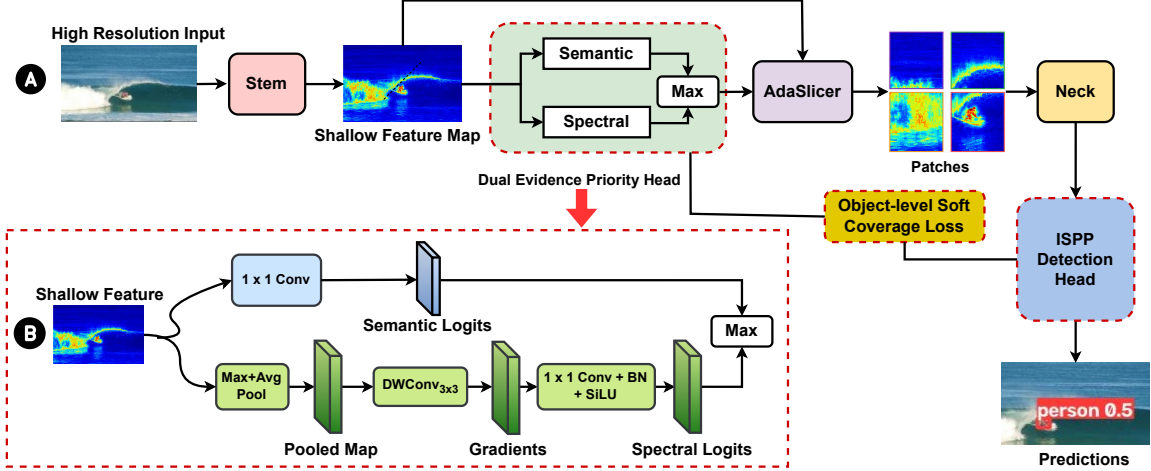


Fig. 2. Overview of HESOD: (a) End-to-end selective pipeline; (b) Dual-Evidence Priority Head. Red dashed boxes highlight our proposed contributions over ESOD.

2. RELATED WORK

2.1. Small Object Detection in High-Resolution Vision

Detecting tiny targets in high-resolution aerial and maritime imagery is severely challenged by extreme scale disparity, low contrast, and massive background areas ($>70\%$ – 80%) [1–3, 12]. Classical approaches utilize density-guided cropping (ClusDet [13], DMNet [14]) or scale-adaptive label assignment (NWD [15], RFLA [16]). Recently, ResBiDet [17] introduced a coarse-to-fine framework using global heatmap guidance for region cropping. However, it relies on multi-scale image-level slicing and boundary stitching. In contrast, HESOD performs early feature-level selective routing directly within the detector trunk and explicitly optimizes instance coverage under irreversible pruning.

2.2. Selective and Dynamic Computation

Dynamic computation accelerates inference by allocating compute strictly to prospective foreground regions (e.g., AutoFocus [5], QueryDet [6], CEASC [7], and sparse location activation [18]). Recently, coarse-to-fine detectors such as ESOD [4] and OSDT [19] predict early spatial priority heatmaps to extract candidate patches. However, existing selectors rely exclusively on a single semantic score supervised by independent per-cell BCE. Under extreme scale disparity, micro targets suffer from semantic feature collapse after downsampling; meanwhile, per-cell BCE lacks joint instance-level coverage constraints, causing fatal false negatives under an asymmetric error cost. HESOD resolves this by combining semantic and spectral dual cues with object-level soft-coverage optimization.

2.3. Spectral Saliency and Frequency Methods

High-frequency edge gradients persist for micro targets even as photometric semantics fade; recent methods exploit this via dense frequency transforms [8], frequency-gradient networks [10], dual-domain alignment [9], or multi-scale compensation [20], but all apply spectral cues *downstream*, incurring substantial latency. In sharp contrast, HESOD repurposes high-frequency gradient saliency *upstream* as an ultra-lightweight routing cue: channel pooling compresses stem features to 2 channels, enabling directional Laplacian and Sobel gradient filtering at constant width, at negligible overhead.

3. PROPOSED METHOD

The architecture of HESOD is shown in Fig. 2. From input $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, a shallow stem extracts dense features $\mathbf{F} \in \mathbb{R}^{H_s \times W_s \times C}$. The *Dual-Evidence Spatial Selector* then coordinates: (i) a priority head fusing semantic objectness with channel-pooled spectral saliency (Fig. 2(b)) into spatial priority map $\mathbf{S} \in [0, 1]^{H_s \times W_s}$; and (ii) an AdaSlicer module (unchanged from ESOD [4]) clustering candidate foreground cells into bounding patches $\mathcal{P}$. Downstream backbone stages and lightweight decoupled heads evaluate only retained patches $\mathcal{P}$, trained under the soft-coverage supervision (Sec. 3.2) via the two-stage protocol (Sec. 3.3).

3.1. Dual-Evidence Priority Estimation

To rescue micro targets with degraded semantic activations, HESOD couples high-level semantic confidence with low-level structural gradient saliency at negligible compute overhead (Fig. 2(b)).

Let $\mathbf{f}_i \in \mathbb{R}^C$ denote the feature vector at spatial location i in $\mathbf{F}$. The semantic branch predicts class-specific logits via a 1×1 convolution: $\mathbf{z}_i^{\text{sem}} = \text{Conv}_{1 \times 1}(\mathbf{f}_i) \in \mathbb{R}^{N_c}$. In parallel, the spectral branch extracts high-frequency structural gradients by first compressing channels via max- and mean-pooling:

$$\mathbf{F}_{\text{pooled}} = \left[\max_c \mathbf{F}_{:::,c} \parallel \frac{1}{C} \sum_{c=1}^C \mathbf{F}_{:::,c} \right] \in \mathbb{R}^{H_s \times W_s \times 2}, \quad (1)$$

where $\mathbf{F}_{:::,c} \in \mathbb{R}^{H_s \times W_s}$ is the c -th channel of $\mathbf{F}$ and $\parallel$ denotes channel-wise concatenation. Compressing $\mathbf{F}$ to 2 channels retains peak edge activation and background energy, after which a learnable depthwise 3×3 convolution bank—initialized with discrete Laplacian and directional Sobel operators—filters $\mathbf{F}_{\text{pooled}}$ into 6 structural channels at $\mathcal{O}(H_s W_s)$ cost independent of width C . In the 2D spatial frequency domain, the discrete Laplacian is approximately isotropic in the low-frequency limit, with $|H(\omega)| \propto \|\omega\|^2$, naturally amplifying edge transitions. Because micro targets ($<16^2$ px) manifest as localized high-frequency spatial impulses whose gradient energy resists downsampling [10], this branch supplies a constant-width prior independent of deep semantic capacity. A 1×1 projection maps features to N_c , yielding spectral logits $\mathbf{z}_i^{\text{spec}} \in \mathbb{R}^{N_c}$.

To prevent confidence dilution on faint targets, evidence fusion enforces *evidence preservation*. Linear combinations risk destructive interference: an unconfident or negative activation in one branch

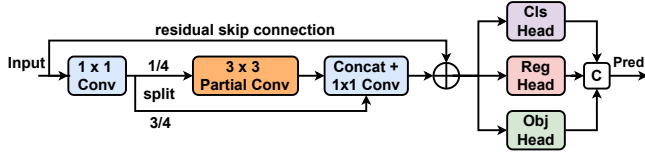


Fig. 3. Inverse-Residual Shared-Parameter Partial-Convolution (ISPP) Head (⊕ denotes channel concatenation).

can drag down a confident signal in the other below the routing threshold. Instead, we formulate a parameter-free elementwise maximum:

$$z_{i,c} = \max(z_{i,c}^{\text{sem}}, z_{i,c}^{\text{spec}}), \quad s_i = \max_{c=1}^{N_c} \sigma(z_{i,c}), \quad (2)$$

This operation functions as an evidence-preserving OR gate: since $z_{i,c} \geq z_{i,c}^{\text{sem}}$ and $z_{i,c} \geq z_{i,c}^{\text{spec}}$, neither branch can ever suppress genuine candidate cues. Cells with priority $s_i > \tau$ are clustered into bounding patches $\mathcal{P}$ for downstream detection.

3.2. Object-Level Soft-Coverage Loss

Standard pixel-wise binary cross-entropy (BCE) evaluates grid cells independently, failing to guarantee instance-level coverage: a selector may over-allocate cells to large objects while dropping micro objects whose receptive fields contain only weak responses. To prevent premature false negatives, we formulate a differentiable object-level soft-coverage loss.

Let $\mathcal{N}(j)$ denote the set of selector cells overlapping ground-truth box $j \in \{1, \dots, N_{\text{gt}}\}$. We formulate a noisy-OR differentiable coverage surrogate and its weighted instance loss:

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i), \quad \ell_j = -w_j \log(p_j^{\text{cover}} + \epsilon), \quad (3)$$

where $w_j = \text{clip}(4/a_j, 1, 5)$ is a scale-adaptive weight based on bounding box area a_j in cells, penalizing misses on micro targets ($a_j < 4$). For any cell $i \in \mathcal{N}(j)$, the gradient of ℓ_j with respect to cell score s_i is:

$$\frac{\partial \ell_j}{\partial s_i} = -\frac{w_j}{p_j^{\text{cover}} + \epsilon} \prod_{k \in \mathcal{N}(j) \setminus \{i\}} (1 - s_k). \quad (4)$$

When neighboring candidate cells have low scores ($s_k \rightarrow 0$), the product term approaches 1, generating a strong gradient to boost cell i and encouraging at least one cell to activate per instance. The overall soft-coverage loss averages over instances: $\mathcal{L}_{\text{cover}} = \frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} \ell_j$. The multi-task training objective is $\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{sel}}(\mathcal{L}_{\text{BCE}} + \lambda_{\text{cov}} \mathcal{L}_{\text{cover}})$, where $\mathcal{L}_{\text{BCE}}$ is the per-cell selector BCE over scores s_i , with $\lambda_{\text{sel}} = 0.2$ and $\lambda_{\text{cov}} = 0.5$.

3.3. Lightweight Decoupled Head and Training Protocol

While dual-evidence routing substantially improves patch coverage by retaining faint foreground regions, evaluating more candidate patches increases downstream detector computation. To recover this overhead without sacrificing selector gains, HESOD incorporates an ISPP decoupled head [11] as a dedicated compute compensator.

As shown in Fig. 3, the ISPP head expands neck feature channels of width C_1 to $2C_1$, partitioning them into an active branch (0.25 ratio) processed by a 3×3 Partial Convolution (PConv) and an identity passthrough branch (0.75 ratio). After linear projection and residual addition, decoupled 1×1 heads predict classification, box regression, and objectness independently, lowering arithmetic complexity while eliminating inter-task gradient interference.

Two-Stage Training Protocol. To ensure stable routing behavior, HESOD employs a two-stage training scheme: (i) *Stage 1*

Table 1. Benchmark results on UAVDT and SeaPerson.

Method	mAP@.5	Params (M)	GFLOPs	Total Recall	BPR	FPS
UAVDT Benchmark (1280 × 1280)						
Faster R-CNN [21]	0.336	43.3	772.5	79.68%	—	18.8
RetinaNet [22]	0.296	36.4	360.9	83.44%	—	19.8
QueryDet [†] [6]	0.257	39.3	— [†]	41.89%	—	21.1
ESOD [4]	<u>0.385</u>	<u>35.9</u>	68.2	<u>85.17%</u>	<u>0.884</u>	117.8
HESOD (Ours)	0.394	26.0	<u>74.9</u>	88.83%	0.940	<u>106.1</u>
Δ vs. ESOD	+0.9 pp	-27.5%	+9.8%	+3.66 pp	+5.6 pp	-9.9%
SeaPerson Benchmark (2048 × 2048)						
Faster R-CNN [21]	0.551	43.3	1546.8	67.27%	—	23.1
RetinaNet [22]	0.473	36.4	942.7	65.98%	—	28.0
QueryDet [†] [6]	0.627	39.3	— [†]	61.56%	—	14.0
ESOD [4]	<u>0.750</u>	<u>35.8</u>	202.4	<u>84.42%</u>	<u>0.947</u>	85.7
HESOD (Ours)	0.773	25.9	<u>208.5</u>	87.75%	0.991	<u>80.6</u>
Δ vs. ESOD	+2.3 pp	-27.6%	+3.0%	+3.33 pp	+4.4 pp	-6.0%

BPR: Best Possible Recall for objects' bounding boxes (BPR^{box}), following ESOD [4]. **Bold**: best; underline: second-best (selective models).

[†]QueryDet uses CSQ (sparse FLOPs unamenable to static profiling; dense query counterparts: 973.5 / 2539.0 GFLOPs at 12.5 / 10.6 FPS on UAVDT / SeaPerson).

(*Selector Convergence*): Train the Dual-Max selector with a coupled head under $\mathcal{L}_{\text{cover}}$, establishing a well-calibrated priority heatmap; and (ii) *Stage 2 (Head Fine-Tuning)*: Freeze the upstream trunk (backbone, dual branches, router) and fine-tune the lightweight ISPP neck and decoupled heads on the frozen routing distribution. Relative to the uncompressed Dual-Max selector, this lightweight head slashes GFLOPs by 16.9% and parameters by 27.5% (on UAVDT) while fully preserving peak selector coverage (0.940), maintaining competitive real-time FPS with only fractional FLOPs overhead vs. ESOD.

4. EXPERIMENTS

4.1. Experimental Setup

We evaluate HESOD on two demanding high-resolution small-object benchmarks: **UAVDT** [2] (1280 × 1280 px, urban vehicle surveillance across 3 classes, 19.9% extremely tiny targets <16² px) and **SeaPerson** [3, 12] (2048 × 2048 px, maritime search-and-rescue with >85% targets <32 px). Both benchmarks use full-resolution images without artificial cropping.

Selective detectors (ESOD family and our ablations) employ YOLOv5m backbones in PyTorch 2.8 trained for 50 epochs under selective protocols [4]. Dense and query baselines (Faster R-CNN, RetinaNet, QueryDet) retain canonical ResNet-50-FPN backbones, evaluated on identical test splits and resolutions. All models are benchmarked on an NVIDIA RTX 5090 GPU. Metrics include COCO mAP@.5 and mAP@.5 : .95, scale-stratified recall across four bins (Very Tiny <16², Tiny 16²–32², Small 32²–96², Med/Large >96² px), and, following ESOD [4], Best Possible Recall for objects' bounding boxes (BPR^{box}), measuring the fraction of ground-truth targets covered (>50% overlap) by selected patches.

4.2. Main Benchmark Comparisons

As shown in Table 1, traditional dense detectors suffer from severe efficiency bottlenecks on high-resolution images: Faster R-CNN only achieves 18.8/23.1 FPS on UAVDT/SeaPerson with GFLOPs as high as 772.5/1546.8; QueryDet [6] is limited by its dynamic query mechanism, with total recall dropping to only 41.89% on UAVDT. HESOD outperforms all baselines: on UAVDT, mAP@.5 is 5.8, 9.8, and 13.7 pp higher than Faster R-CNN, RetinaNet, and

Table 2. Module ablation, efficiency, and scale recall breakdown on UAVDT and SeaPerson.

Configuration	Detection Accuracy and Computational Efficiency						Physical-Scale Recall Breakdown				
	mAP @.5	mAP @.5:.95	BPR	Params (M)	GFLOPs	FPS	Very Tiny ($< 16^2$)	Tiny (16^2-32^2)	Small (32^2-96^2)	Med/Lrg ($> 96^2$)	Total Recall
UAVDT Benchmark (1280 × 1280)							(74,375)	(206,015)	(86,282)	(7,325)	(373,997)
(1) ESOD Baseline*	0.385	0.214	0.884	35.9	68.2	117.8	79.43%	84.21%	94.54%	59.78%	85.17%
(2) Semantic-only	0.384	<u>0.217</u>	0.906	35.9	75.0	<u>113.8</u>	79.13%	83.91%	94.00%	59.95%	84.82%
(3) Spectral-only	0.387	0.209	<u>0.942</u>	36.0	93.7	102.3	82.43%	88.72%	96.66%	<u>66.80%</u>	88.87%
(4) (3) + Channel-Pooled	0.394	0.209	0.963	35.9	99.3	97.6	86.88%	92.23%	97.74%	66.43%	91.93%
(5) Dual-Concat	0.371	0.205	0.919	35.9	83.9	107.2	79.09%	85.85%	95.85%	62.66%	86.35%
(6) Dual-Max	0.395	0.218	0.940	35.9	90.1	102.3	<u>85.69%</u>	<u>89.97%</u>	<u>97.37%</u>	65.99%	<u>90.36%</u>
(7) HESOD (Ours)	0.394	0.215	0.940	26.0	<u>74.9</u>	106.1	80.76%	88.98%	97.11%	69.16%	88.83%
SeaPerson Benchmark (2048 × 2048)							(82,417)	(174,816)	(42,987)	(155)	(300,375)
(1) ESOD Baseline*	0.750	0.320	0.947	35.8	202.4	85.7	74.03%	86.78%	94.74%	79.35%	84.42%
(2) Semantic-only	0.769	0.325	0.991	35.8	266.8	73.5	75.49%	91.57%	95.71%	81.94%	87.75%
(3) Spectral-only	0.770	<u>0.327</u>	0.992	35.8	267.4	71.7	75.23%	<u>91.69%</u>	95.89%	<u>86.45%</u>	87.77%
(4) (3) + Channel-Pooled	0.767	0.324	0.988	35.8	263.4	72.7	76.13%	91.36%	95.50%	87.10%	87.77%
(5) Dual-Concat	0.772	0.326	0.986	35.8	281.2	69.8	<u>76.00%</u>	91.50%	95.68%	80.00%	<u>87.84%</u>
(6) Dual-Max	0.778	0.330	<u>0.991</u>	35.8	255.1	73.8	75.65%	92.06%	<u>95.85%</u>	81.94%	88.10%
(7) HESOD (Ours)	<u>0.773</u>	<u>0.327</u>	<u>0.991</u>	25.9	<u>208.5</u>	80.6	75.71%	91.47%	95.72%	80.65%	87.75%

*Baseline uses BCE only; others add $\mathcal{L}_{\text{cover}}$ on top. Parenthetical values denote ground-truth counts per bin. One-to-one matching at conf ≥ 0.001 , IoU ≥ 0.5 . **Bold**: best; underline: second-best.

QueryDet respectively at $5.0\times-5.6\times$ higher FPS (106.1 vs. 18.8–21.1); on SeaPerson, mAP@.5 exceeds all baselines by 14.6~30.0 pp at $2.9\times-5.8\times$ higher FPS, with Very Tiny recall reaching 75.71% (+27.0 pp over QueryDet).

Against the same-architecture selective baseline ESOD [4]: on UAVDT, mAP@.5 increases from 0.385 to 0.394 (+0.9 pp), BPR from 0.884 to 0.940 (+5.6 pp), total recall by +3.66 pp, with bus class recall jumping from 67.7% to 81.4% (+13.7 pp); parameters are reduced from 35.9M to 26.0M (−27.5%) at 106.1 FPS. On SeaPerson, mAP@.5 increases from 0.750 to 0.773 (+2.3 pp), BPR reaches 0.991 (+4.4 pp), total recall increases by +3.33 pp, successfully recalling more than 10,000 micro maritime targets missed by ESOD; GFLOPs only increase by 3.0%, parameters are compressed by 27.6% at 80.6 FPS. The modest GFLOPs overhead stems from the selector routing more genuine target regions downstream rather than computational redundancy; HESOD controls this within 10% via the lightweight head design while maintaining high patch coverage (94.0%/99.1% BPR on UAVDT/SeaPerson).

4.3. Ablation Studies and Scale Analysis

We conduct controlled ablation experiments on the ESOD baseline by adding components incrementally, with results in Table 2.

Dual-Evidence Fusion. Using only the semantic branch yields 0.906 BPR and 84.82% total recall on UAVDT, showing limited improvement due to semantic feature collapse during downsampling; the spectral-only branch achieves stronger BPR of 0.942 and 88.87% total recall, as gradient features remain robust on micro targets. Comparing fusion strategies, linear concatenation (config. 5) achieves only 0.371 mAP@.5—lower than either single branch—because weak semantic responses introduce destructive interference that suppresses confident spectral activations. In contrast, element-wise maximum (Dual-Max, config. 6) implements an evidence-preserving OR gate, achieving 0.395/0.778 mAP@.5 and 0.940/0.991 BPR on UAVDT/SeaPerson, the best among all single-stage selector configurations.

Soft-Coverage Loss. The ESOD baseline trained with per-pixel BCE achieves only 0.884 BPR and 79.43% Very Tiny recall on UAVDT. With the proposed $\mathcal{L}_{\text{cover}}$, the semantic branch’s Very Tiny recall is flat at 79.13%, while the spectral branch’s rises to 82.43%; under Dual-Max fusion it reaches 85.69% (+6.26 pp). The noisy-OR formulation provides strong gradients to low-scoring covering

cells when a target is under-covered, encouraging at least one cell to exceed the routing threshold and reducing unrecoverable missed detections.

ISPP Head Efficiency. Without the ISPP head, Dual-Max incurs 90.1 GFLOPs and 102.3 FPS on UAVDT. Adding ISPP reduces GFLOPs by 16.9% to 74.9, boosts FPS to 106.1, and compresses parameters by 27.5%, with mAP@.5 decreasing by only 0.001 and BPR unchanged at 0.940. On SeaPerson, ISPP cuts GFLOPs by 18.2% and parameters by 27.6%, lifting FPS from 73.8 to 80.6 with negligible accuracy loss (0.778 → 0.773 mAP).

Channel Pooling. Without channel pooling, full-width spectral filtering limits batch size to 2 on SeaPerson due to memory constraints, yielding only 0.763 mAP@.5. Compressing features to 2 channels enables batch-8 training, boosting mAP@.5 by +1.5 pp to 0.778, reducing GFLOPs by 8.2%, and further lifting Very Tiny recall by +1.2 pp.

Error Bottleneck Diagnosis. Coverage-loss supervision alone reduces selector false negatives from 25.6% to 19.2% on SeaPerson’s semantic branch (BPR 0.947→0.991), while HESOD adds more than 13,600 detections on UAVDT (total recall 85.17%→88.83%), with BPR rising from 0.884 to 0.940, shifting the bottleneck to downstream regression.

Latency Profiling. On an RTX 5090 GPU, stem extraction takes 1.2 ms, selector 1.8 ms, AdaSlicer 0.9 ms, and ISPP 5.5 ms, achieving 106.1/80.6 FPS on UAVDT/SeaPerson respectively, demonstrating real-time throughput under this evaluation setup.

5. CONCLUSION

We presented HESOD, uniting semantic objectness with constant-width spectral saliency via monotonic max-fusion to address early semantic collapse on micro targets. An object-level soft-coverage loss ($\mathcal{L}_{\text{cover}}$) optimizes instance coverage under asymmetric pruning risk, while a lightweight ISPP head amortizes downstream overhead without sacrificing selector gains. On UAVDT and SeaPerson, HESOD improves mAP@.5 by 0.9 pp and 2.3 pp (total recall +3.66 pp and +3.33 pp), cuts parameters by >27%, and achieves 106.1/80.6 FPS on an RTX 5090. Dual-evidence routing reaches 0.940/0.991 BPR, cutting selector false negatives to 19.2% on SeaPerson and shifting the bottleneck to downstream regression. These results confirm that spectral structural priors enable accuracy-efficiency co-design for resource-efficient deployment.

6. REFERENCES

- [1] Pengfei Zhu, Longyin Wen, Dawei Du, Xiao Bian, Heng Fan, Qinghua Hu, and Haibin Ling, "Detection and tracking meets drones: A benchmark," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 7388–7404, 2021.
- [2] Dawei Du, Yuankai Qi, Hongyang Yu, Yifan Yang, Kaiwen Duan, Guorong Li, Weigang Zhang, Qingming Huang, and Ling Shao, "The unmanned aerial vehicle benchmark: Object detection and tracking," in *European Conference on Computer Vision*, 2018, pp. 370–386.
- [3] Xue-Yang Zhang, Guanying Xu, Jin-Yu Fan, and Haibin Ling, "Scale match for tiny person detection," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2020, pp. 1256–1265.
- [4] Kai Liu, Zhunga Fu, Shan Jin, Zhipeng Chen, Feng Zhou, Rong Jiang, Yonghong Chen, and Jinwei Ye, "ESOD: Efficient small object detection on high-resolution images," *IEEE Transactions on Image Processing*, vol. 34, pp. 183–195, 2025.
- [5] Mahyar Najibi, Bharat Singh, and Larry S. Davis, "Auto-Focus: Efficient multi-scale inference," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 9745–9755.
- [6] Chenhongyi Yang, Zehao Huang, and Naiyan Wang, "Query-Det: Cascaded sparse query for accelerating high-resolution small object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 16668–16677.
- [7] Bowei Du, Yanan Huang, Jiahui Chen, and Dong Huang, "Adaptive sparse convolutional networks with global context enhancement for faster object detection on drone images," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 13435–13444.
- [8] Hao Sun, Rui Wang, Yufeng Li, Chenhongyi Yang, Zehao Huang, Naiyan Wang, and Lin Zhou, "SET: Spectral enhancement for tiny object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 1–10.
- [9] Fangrui Guo, Junwei Wu, and Quan Zhang, "Dual-domain feature-guided task alignment for enhanced small object detection," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2025, pp. 1–5.
- [10] M. Zhao et al., "Frequency-gradient collaborative network for infrared small target detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1–14, 2025.
- [11] Xiaocong Xu, Qi Li, Jeng-Shyang Pan, Linfeng Wei, and Xiao-Zhi Gao, "ESOD-YOLO: An enhanced efficient small object detection framework for aerial images," *Computing*, vol. 107, pp. 54, 2025.
- [12] Leon A. Varga, Benjamin Kiefer, Martin Messmer, and Andreas Zell, "SeaDronesSee: A maritime benchmark for detecting humans in open water," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 2260–2270.
- [13] Fanyi Yang, Heng Fan, Peng Chu, Erik Blasch, and Haibin Ling, "Clustered object detection in aerial images," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 8311–8320.
- [14] Changlin Li, Taojiannan Tao, Chen Gong, Xiaojun Chang, Liang-Chieh Chen, and Zizhang Wu, "Density map guided object detection in aerial images," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 190–199.
- [15] Jinwang Wang, Chang Xu, Wen Yang, and Lei Yu, "A normalized gaussian Wasserstein distance for tiny object detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–13, 2021.
- [16] Chang Xu, Jinwang Wang, Wen Yang, Huai Yu, Lei Yu, and Gui-Song Xia, "RFLA: Gaussian receptive field based label assignment for tiny object detection," in *European Conference on Computer Vision*, 2022, pp. 526–543.
- [17] Chen Guo, Yang Li, Jiaman Ma, Zhenyu Fang, Zuoang Niu, and Hongbo Xu, "RESBIDET: Efficient dual-branch small object detection for UAVs under resource-constrained conditions," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026, pp. 3156–3160.
- [18] Yixuan Li et al., "Adaptive sparse feature location activation strategy for sparse detectors on drone images," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2025, pp. 1–5.
- [19] Shigeng Wang et al., "Foreground-enhanced coarse-to-fine detection for UAV small objects," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026, pp. 1–5.
- [20] Kai Wang et al., "High-frequency detail compensation and multi-scale feature fusion net for UAV remote sensing object detection," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026, pp. 1–5.
- [21] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 6, pp. 1137–1149, 2017.
- [22] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2017, pp. 2980–2988.